

An Internship Report

Of

ELECTRIC VEHICLE MODELING

AND

SIMULATION DESIGN VIRTUAL INTERNSHIP

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

**B.Tech (Computer
Science and Engineering)**



By

Roll Number of Student: 2410031537

Name of Student: SHIVAM KUMAR

Batch: 2024-2028

ANDIDATE'S DECLARATION

I, SHIVAM KUMAR, do hereby declare that the internship report titled “Electric Vehicle Modeling And Simulation Design Virtual Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-

Student Name: SHIVAM KUMAR

Roll No.: 2410031537

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their inspiring guidance, invaluable constructive criticism, and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on various aspects related to this internship.

I would like to thank the **AICTE – EduSkills Virtual Internship Program team, EduSkills Academy, and MathWorks** for providing me with the opportunity and a structured platform to learn and gain hands-on exposure in the domain of **Electric Vehicle Modeling and Simulation Design**. I am also thankful to my institute, **IILM University, Greater Noida**, and the faculty of the **School of Computer Science and Engineering** for their continuous support, guidance, and encouragement throughout the internship.

I express my heartfelt gratitude to my **parents for their blessings, constant support, and encouragement**, which motivated me to successfully complete this internship.

Signature:-

Student Name: SHIVAM KUMAR

Roll No.: 2410031537

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3. INTERNSHIP COMPLETION CERTIFICATE



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MINISTRY OF
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NATIONAL
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PORTAL



EduSkills®
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Certificate of Virtual Internship

This is to certify that

shivam kumar

IILM University, Greater Noida

has successfully completed the 8-weeks

Electric Vehicle Modeling and Simulation Design Virtual Internship

During June - August 2026

May this Internship learning propel you toward a bright and successful career.

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Chief Coordinating Officer (CCO)
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Certificate ID : 4ad8bd95e2683dcd3be1

Student ID: STU699ac8fcc6e891771751676



GRADE- O (Outstanding)-90-100 | E (Excellent)-80-89 | A (Very Good)-70-79 | B (Good)-60-69 | C (Fair)-50-59 | D (Average)-40-49 | P (Pass)-30-39 | F (Fail)-Below 30

Internship Offer Letter



Ref No: C17Y2026M081495078

Internship Offer Letter

Student Name : shivam kumar
Stream : B.Tech
Branch : B.Tech(Computer Science and Engineering)
Institute Name : IILM University, Greater Noida
Internship Domain : Electric Vehicle Modeling and Simulation Virtual Internship Program
Internship Duration : June 2026 - August 2026
AICTE Student ID : STU699ac8fcc6e891771751676

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the **8-week Virtual Internship** undertaken under the **AICTE – EduSkills Virtual Internship Program**, in the domain of **“Electric Vehicle Modeling and Simulation Design”**, supported by **MathWorks**. The internship was carried out from **June 2026 to August 2026** as part of the requirement of the **Bachelor of Technology (Computer Science and Engineering)** programme at **IILM University, Greater Noida**.

The primary aim of this internship was to develop a strong practical understanding of **Electric Vehicle (EV) technology, modeling, and simulation** using **MATLAB and Simulink**. The internship provided exposure to important concepts related to electric vehicle systems, including **EV components, battery systems, electric motors, power electronics, vehicle dynamics, and system-level simulation**.

The internship followed a structured learning approach that combined **theoretical concepts with practical exercises, simulations, assignments, and assessments**. Through these activities, I gained practical knowledge of how electric vehicle systems can be modeled and analyzed using simulation-based tools. The internship also helped me understand the importance of simulation in **designing, testing, and improving electric vehicle performance** before implementing systems in real-world applications.

On successful completion of the internship requirements, an **AICTE– EduSkills Virtual Internship Certificate** with an **“Outstanding” Grade (Grade O)** was awarded, supported by **MathWorks** and recognized under the **AICTE and National Internship Portal**.

4.2 Organization Profile

EduSkills Academy is an organization working with the vision of “**Nation Building Through Skills**” and focuses on providing industry-oriented learning and skill development opportunities to students. EduSkills collaborates with the **All India Council for Technical Education (AICTE)** to conduct the **AICTE – EduSkills Virtual Internship Program** under the **AICTE National Internship Portal**.

For the **Electric Vehicle Modeling and Simulation Design** internship track, EduSkills provided a structured learning platform supported by **MathWorks**. The program focused on developing practical knowledge of **Electric Vehicle (EV) technologies, MATLAB, and Simulink**, along with concepts related to EV modeling and simulation.

The internship curriculum was designed to combine **theoretical learning with practical simulation-based activities, assignments, assessments, and hands-on exercises**. Through this program, students gained industry-relevant knowledge and developed an understanding of how simulation tools can be used for the **design, analysis, and performance evaluation of electric vehicle systems**.

4.3 Problem Statement

Although a large number of engineering students learn the basic concepts of **Electric Vehicles, batteries, electric motors, and power systems** during their academic coursework, many students have limited opportunities to gain structured, hands-on experience in **modeling and simulating complete electric vehicle systems** using industry-standard tools such as **MATLAB and Simulink**. Concepts related to EV architecture, battery performance, motor

control, vehicle dynamics, and power management are often understood mainly at a theoretical level, with limited practical exposure to simulation and system analysis.

The problem addressed by this internship was to **practically understand and apply the modeling and simulation workflow for Electric Vehicles** using MATLAB and Simulink. The internship focused on developing knowledge of EV components and their interactions, creating simulation models, analyzing system performance, and understanding how simulation can be used to **design, test, and optimize electric vehicle systems** before their real-world implementation.

4.4 Project Objectives

To understand the **fundamental concepts of Electric Vehicles (EVs)** and their major components.

To learn the basics of **EV modeling and simulation** using **MATLAB and Simulink**.

To understand the working and characteristics of **EV batteries, electric motors, power electronics, and energy management systems**.

To develop practical skills in creating **electric vehicle system models** and performing simulation-based analysis.

To study **vehicle dynamics and performance parameters** such as speed, acceleration, torque, power, and energy consumption.

To analyze the performance of different **EV components and subsystems** through simulation.

To understand the role of **MATLAB and Simulink** in the design, testing, and optimization of electric vehicle systems.

To gain hands-on experience in **simulation-based problem solving** and interpreting simulation results.

To develop an understanding of how EV models can be used to **evaluate and improve vehicle performance and efficiency**.

To successfully complete all the required **internship assignments, assessments, and Internship Grade Point Assessment**.

4.5 Scope of the Project

The scope of this internship covers the complete **Electric Vehicle Modeling and Simulation workflow**, starting from understanding the fundamentals of electric vehicles and their major components, and progressing towards the **modeling, simulation, and analysis of EV systems** using **MATLAB and Simulink**. The internship includes the study of **battery systems, electric motors, power electronics, vehicle dynamics, energy consumption, and overall EV performance**.

The project also focuses on developing and analyzing simulation models to understand the interaction between different EV components and evaluate important performance parameters such as **speed, torque, power, energy consumption, and efficiency**. It provides practical exposure to simulation-based design, testing, and performance analysis of electric vehicle systems.

The internship was limited to a **virtual, simulation and lab-based learning environment** provided through the **AICTE– EduSkills Virtual Internship**

Program, with support from **MathWorks**, and did not involve the deployment of any electric vehicle model or system in a live commercial or industrial environment.

4.6 Technologies and Tools Used

Modeling & Simulation Tools: MATLAB, Simulink

Electric Vehicle Modeling: Electric Vehicle (EV) system models, battery models, electric motor models, and vehicle dynamics models

Battery & Energy Systems: Battery modeling, State of Charge (SOC) analysis, power and energy consumption analysis

Electric Motor & Power Electronics: Electric motor modeling, motor performance analysis, power electronic components and control systems

Simulation & Analysis: MATLAB/Simulink simulation, system-level modeling, performance analysis, and simulation result evaluation

Learning Platform: AICTE– EduSkills Virtual Internship Program learning modules, assignments, assessments, and practical simulation exercises

Supporting Organization: MathWorks, for MATLAB and Simulink-based learning and simulation activities

4.7 System Architecture

The general architecture followed during the internship for the **Electric Vehicle Modeling and Simulation Design** track can be summarized in the following layers:

- **Input Layer:** Provides the required parameters and inputs such as **battery specifications, vehicle speed, motor characteristics, load conditions, and other EV system parameters.**
- **Component Modeling Layer:** Represents the major components of an electric vehicle, including the **battery, electric motor, power electronics, controller, and vehicle dynamics.**
- **Control Layer:** Handles the control and management of the EV system, including **motor control, power management, and battery-related control parameters.**
- **Simulation Layer:** The complete EV model is simulated using **MATLAB and Simulink** to study the behavior and interaction of different components under different operating conditions.
- **Analysis Layer:** Simulation results are analyzed using parameters such as **speed, torque, power, battery State of Charge (SOC), energy consumption, and efficiency.**
- **Optimization Layer:** The simulation results can be used to identify performance issues and improve the **efficiency, stability, and overall performance** of the electric vehicle system.
- **Assessment Layer:** **Assignments, practical simulation exercises, quizzes, and the Internship Grade Point Assessment** were used to evaluate learning progress and successful completion of the internship.

4.8 Methodology

The internship followed a structured, **week-wise methodology** combining conceptual learning, hands-on simulation exercises, practical activities, and

assessment-based learning. The internship focused on **Electric Vehicle Modeling and Simulation Design** using MATLAB and Simulink. The learning process progressed from basic EV concepts to component modeling, system simulation, performance analysis, and final assessment, as summarized below.

Week	Module	Description
1	Introduction to Electric Vehicles	Introduction to EVs, their working principles, types, major components, advantages, and basic EV architecture.
2	EV Components and Systems	Study of important EV components including batteries, electric motors, power electronics, controllers, and charging systems.
3	Battery Modeling and Analysis	Understanding battery characteristics, battery models, State of Charge (SOC), voltage, current, power, and energy-related parameters.
4	Electric Motor Modeling	Study and modeling of electric motors used in EVs, along with analysis of motor speed, torque, power, and operating characteristics.
5	EV Powertrain Modeling	Development and understanding of an EV powertrain model by integrating the battery, motor, controller, and other major subsystems.
6	Vehicle Dynamics and Simulation	Modeling vehicle dynamics and analyzing parameters such as vehicle speed, acceleration, torque, power demand, and energy consumption.

Week	Module	Description
7	EV System Simulation and Performance Analysis	Performing simulations using MATLAB and Simulink and analyzing the overall performance, efficiency, and behavior of the EV system under different conditions.
8	Advanced EV Modeling and Simulation	Applying the concepts learned to improve and analyze EV models through simulation-based activities and practical exercises.
Final	Final Assessment / Internship Grade Point Assessment	Completion of the final assessment and evaluation of the knowledge and practical skills gained throughout the internship.

Note: There was no stipend associated with this internship.

Each week consisted of **learning modules, practical simulation activities, assignments, and assessments** to evaluate conceptual understanding and practical knowledge. In the final phase, an **Internship Grade Point Assessment** was conducted, and the overall internship performance was evaluated based on the completion of the weekly learning activities and final assessment.

4.9 Expected Outcomes

- Ability to explain the **basic concepts and working principles of Electric Vehicles (EVs)** and their major components.
- Practical understanding of **EV batteries, electric motors, power electronics, controllers, and vehicle powertrain systems**.

- Hands-on experience in developing **Electric Vehicle models using MATLAB and Simulink.**
- Ability to understand and analyze important EV parameters such as **speed, torque, power, State of Charge (SOC), energy consumption, and efficiency.**
- Practical knowledge of performing **simulation and performance analysis** of electric vehicle systems under different operating conditions.
- Ability to interpret simulation results and identify opportunities for improving **EV performance and energy efficiency.**
- Development of **simulation-based problem-solving skills** and familiarity with industry-oriented EV modeling tools.
- Successful completion of the **weekly learning modules, practical exercises, assignments, and Internship Grade Point Assessment,** resulting in an **“Outstanding” grade.**

4.10 Certificates of Completion and Communication Proof

The Internship Offer Letter and the Internship Completion Certificate issued by EduSkills Academy, confirming successful completion of the 8-week AICTE – EduSkills Virtual Internship Program in Google AI-ML, are attached in Chapter 3 of this report. The official completion e-mail from EduSkills Foundation.

5. BIBLIOGRAPHY/REFERENCES

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